# C++ Lab: Classes, Objects and Constructors

Write, compile and run a C++ program for each problem. The examples show the expected results; the exact wording of your output may differ.

## 1. Rectangle

Write a program to find the area and perimeter of a rectangle. Use a class so that the measurements and the calculations are kept together.

**Data members:** length and breadth, both of type double.

**Member functions:** input() reads the length and breadth. area() returns length × breadth. perimeter() returns 2 × (length + breadth).

**In main():** Create one Rectangle object, call input(), and display the area and perimeter.

**Example input:** Length = 6, Breadth = 4

**Expected output:** Area = 24, Perimeter = 20

## 2. Student

Write a program to store the details of a student and calculate the total and average marks in three subjects.

**Data members:** name (string), rollNo (int), and marks1, marks2 and marks3 (int).

**Member functions:** input() reads all the details. total() returns the sum of the three marks. average() returns the total divided by 3.0. display() shows the name, roll number, total and average.

**In main():** Create one Student object, call input(), then call display().

**Example input:** Name = Sana, Roll number = 101, Marks = 80, 90, 85

**Expected output:** Name: Sana, Roll number: 101, Total: 255, Average: 85

## 3. Temperature

Write a program to convert a temperature from Celsius to Fahrenheit. Use the formula F = (C × 9 / 5) + 32.

**Data members:** celsius, of type double.

**Member functions:** input() reads the temperature in Celsius. toFahrenheit() returns the converted value. display() shows both temperatures.

**In main():** Create one Temperature object, call input(), then call display().

**Example input:** Celsius = 37

**Expected output:** 37 Celsius = 98.6 Fahrenheit

## 4. Bank account

Write a program for a simple bank account. The balance must be private, so that it can change only through the public member functions.

**Data members:** holderName (string) and balance (double), both private.

**Member functions:** setAccount() sets the name and a starting balance of 0. deposit() adds an amount to the balance, and only accepts an amount greater than 0. withdraw() subtracts an amount, and prints "Insufficient balance" if the amount is more than the balance. showBalance() displays the balance.

**In main():** Create one object, deposit 500, withdraw 200, try to withdraw 1000, and display the balance.

**Example input:** Deposit 500, Withdraw 200, Withdraw 1000

**Expected output:** Insufficient balance. Balance = 300

## 5. Shopping bill

Write a program to calculate the bill for an item bought from a shop.

**Data members:** name (string), price (double) and quantity (int).

**Member functions:** input() reads the item details. totalBill() returns price × quantity. display() shows the item name, price, quantity and total bill.

**In main():** Create one Product object, call input(), then call display().

**Example input:** Item = Notebook, Price = 45, Quantity = 4

**Expected output:** Notebook: 45 × 4, Total bill = 180

## 6. Default constructor

Write a program to show that a default constructor runs automatically when an object is created. A default constructor takes no arguments.

**Data members:** title (string) and price (double).

**Member functions:** The default constructor Book() sets the title to "Unknown" and the price to 0. display() shows the title and price.

**In main():** Create one Book object without passing any values, then call display(). Do not call the constructor yourself.

**Example input:** None

**Expected output:** Title: Unknown, Price: 0

## 7. Parameterized constructor

Write a program to set the radius of a circle at the moment the object is created, by passing it to a constructor. Use 3.14 as the value of pi.

**Data members:** radius, of type double.

**Member functions:** The constructor Circle(double r) stores r in radius. area() returns 3.14 × radius × radius.

**In main():** Create two objects with radius 7 and radius 3.5, and display the area of each circle.

**Example input:** Radius = 7 and Radius = 3.5

**Expected output:** Area = 153.86 and Area = 38.465

## 8. Constructor overloading

Write a program in which a class has more than one constructor. The compiler chooses the constructor by matching the arguments passed.

**Data members:** length, breadth and height, all of type double.

**Member functions:** The default constructor Box() sets all three sides to 1. The constructor Box(double l, double b, double h) sets the sides to the values passed. volume() returns length × breadth × height.

**In main():** Create one object with no values and one with the values 2, 3 and 4. Display both volumes.

**Example input:** No values, then 2, 3, 4

**Expected output:** Volume = 1 and Volume = 24

## 9. Copy constructor

Write a program to create a new object as a copy of an existing object. A copy constructor takes an object of the same class as its argument.

**Data members:** x and y, both of type int.

**Member functions:** The constructor Point(int a, int b) sets x and y. The copy constructor Point(const Point &p) copies x and y from p. display() shows the point in the form (x, y).

**In main():** Create a point with the values 3 and 4. Create a second point as a copy of the first, and display both points.

**Example input:** x = 3, y = 4

**Expected output:** Original: (3, 4), Copy: (3, 4)

## 10. Array of objects

Write a program to store the details of three cars in an array of objects, and display them using a loop.

**Data members:** model (string) and price (int).

**Member functions:** input() reads the model and price of one car. display() shows the model and price of one car.

**In main():** Create an array of three Car objects. Use one loop to call input() for each object, and a second loop to call display().

**Example input:** Swift 700000, Creta 1100000, Thar 1500000

**Expected output:** Swift: 700000, Creta: 1100000, Thar: 1500000